

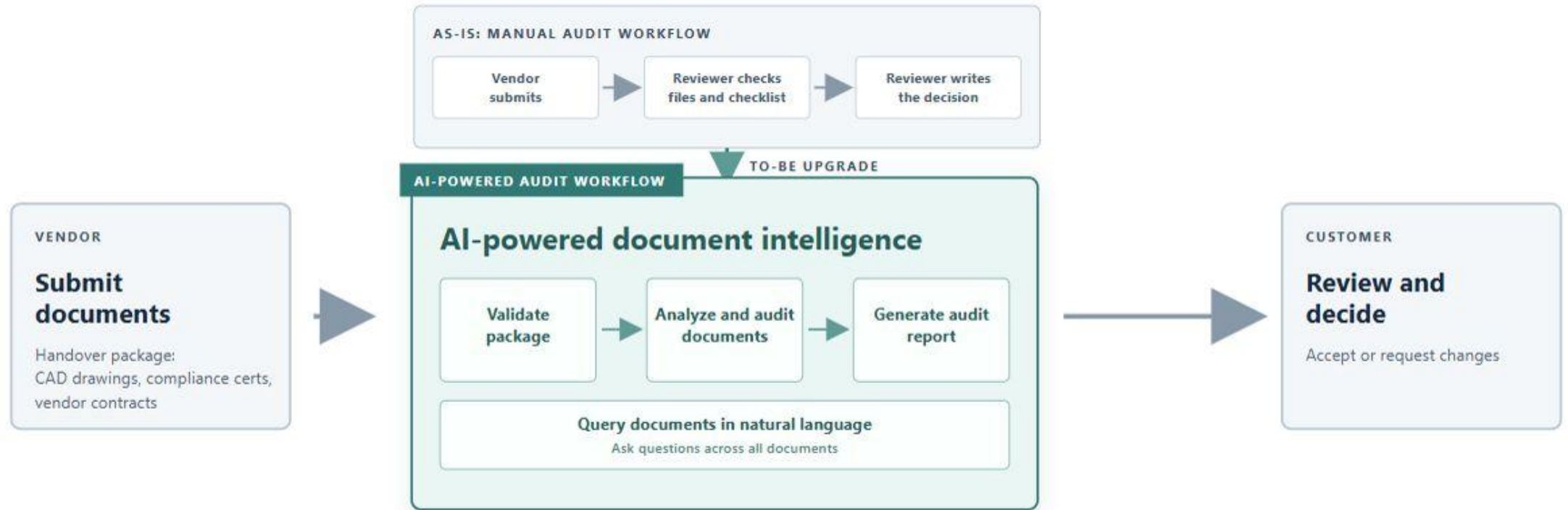
AI Document Audit & Knowledge Assistant System

Team 6

Agenda

1. Product Overview
2. Architecture Decisions
3. System Architecture
4. Infrastructure Architecture
5. Security, Observability, and Scaling
6. AWS Fit (or other providers)
7. Sizing & Cost

Product Overview



Architecture Decisions (1)

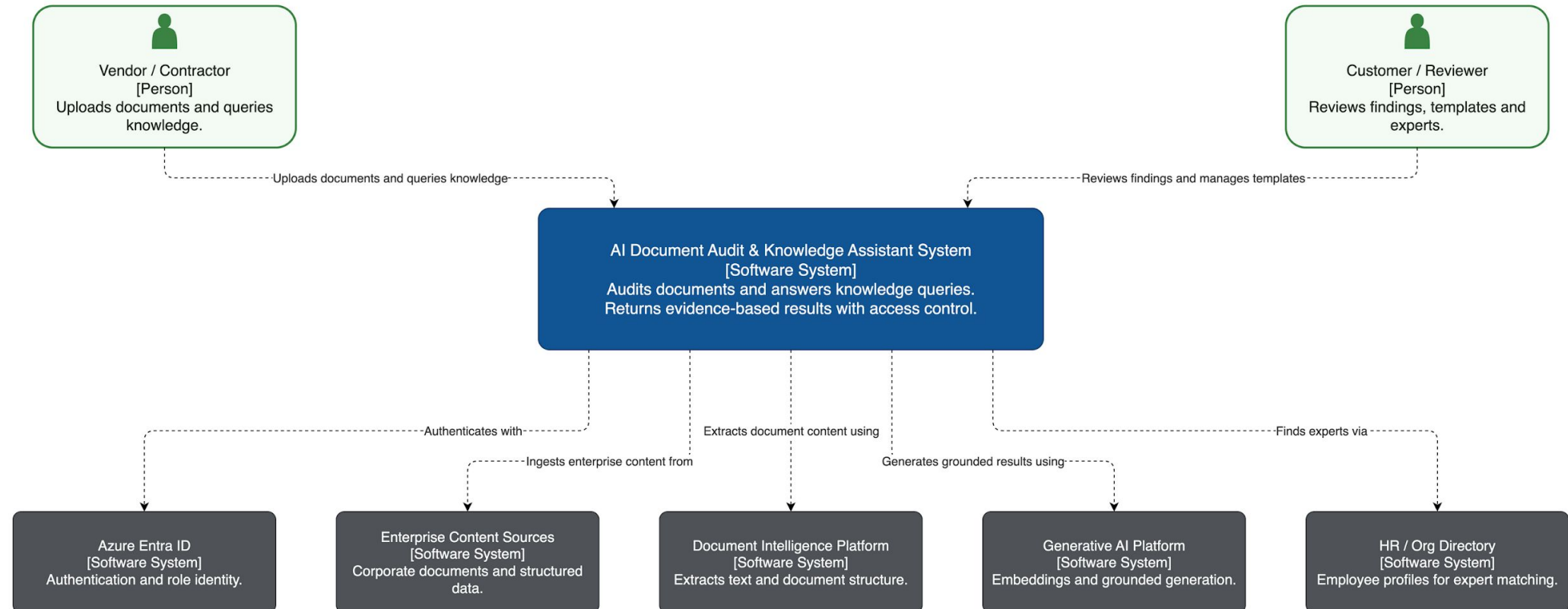
- **Cloud Platform:** Use Azure because the customer already runs their entire system on this platform, and deploy across multiple Availability Zones in to ensure availability if one zone fails.
- **Technology Stack:** Use Python with FastAPI for the backend because it supports rapid API development, asynchronous processing, and easy integration with AI and ETL services. Use ReactJS for the frontend to build a responsive, component-based interface with reusable UI elements and efficient state management.
- **Identity Platform:** Use Microsoft Entra ID for managed identities because the customer already uses it as their identity platform. This avoids introducing another identity system while simplifying access management.
- **Secure Web Entry Path:** Use layered edge protection, web application firewalling, and private application access to reduce threats and access-point failures.
- **Microservices:** Separate services so each can scale, deploy, and recover independently.

Architecture Decisions (2)

- **Asynchronous ETL Pipeline:** Use Celery to distribute long-running ETL tasks across worker processes, enabling independent scaling, automatic retries, task monitoring, and failure isolation. Combine it with idempotent tasks and a dead-letter queue to prevent processing failures from causing data loss.
- **Resilient Data Storage:** Enable high availability (HA) to keep documents and audit records accessible during hardware, zone, or service failures. Use redundant storage, regular database backups, and point-in-time recovery to minimize downtime and prevent data loss.
- **Private Networking:** Use private connectivity for internal services and data stores to keep traffic off the public Internet and reduce the attack surface.
- **Managed Document Understanding:** Use a managed document-understanding service to reduce the need to build and maintain complex OCR, layout-analysis, and document-classification capabilities in-house. It provides scalable, continuously maintained processing for different document formats and improves extraction accuracy.
- **Human-in-the-Loop AI Decisions:** Use AI to generate audit findings, while requiring customer review before final decisions are made.

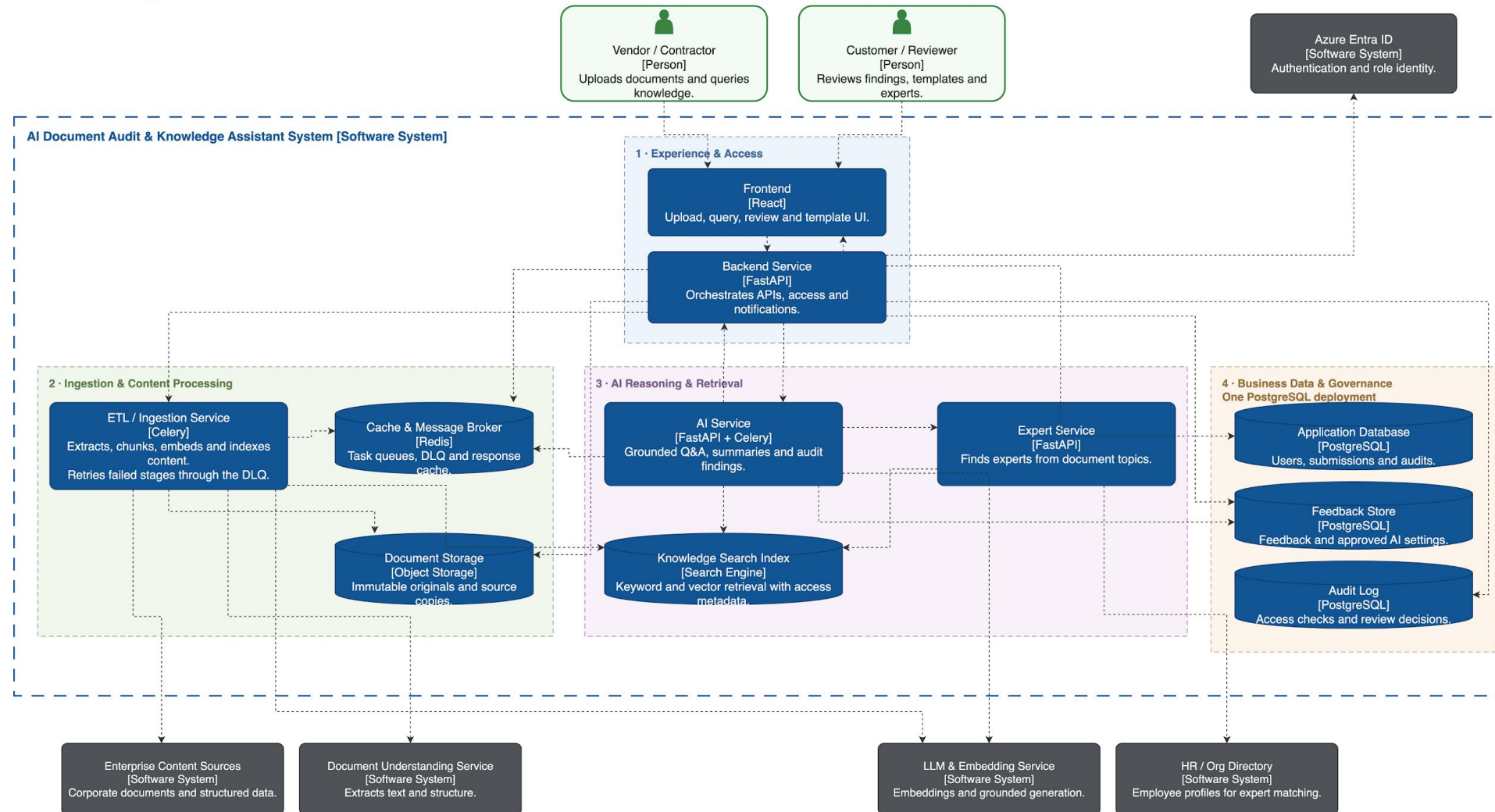
System Architecture - System Context

System Context Diagram

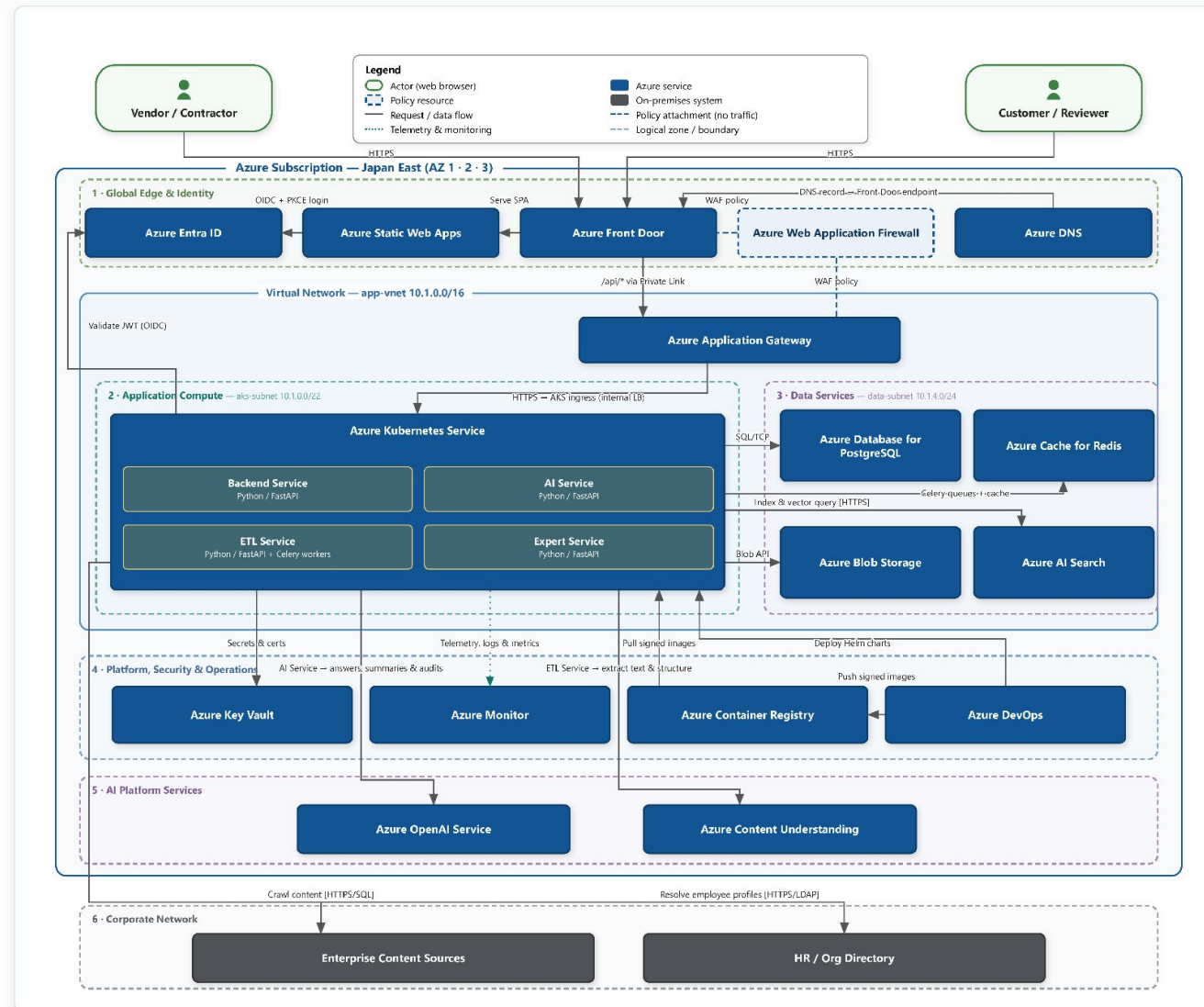


System Architecture - Container View

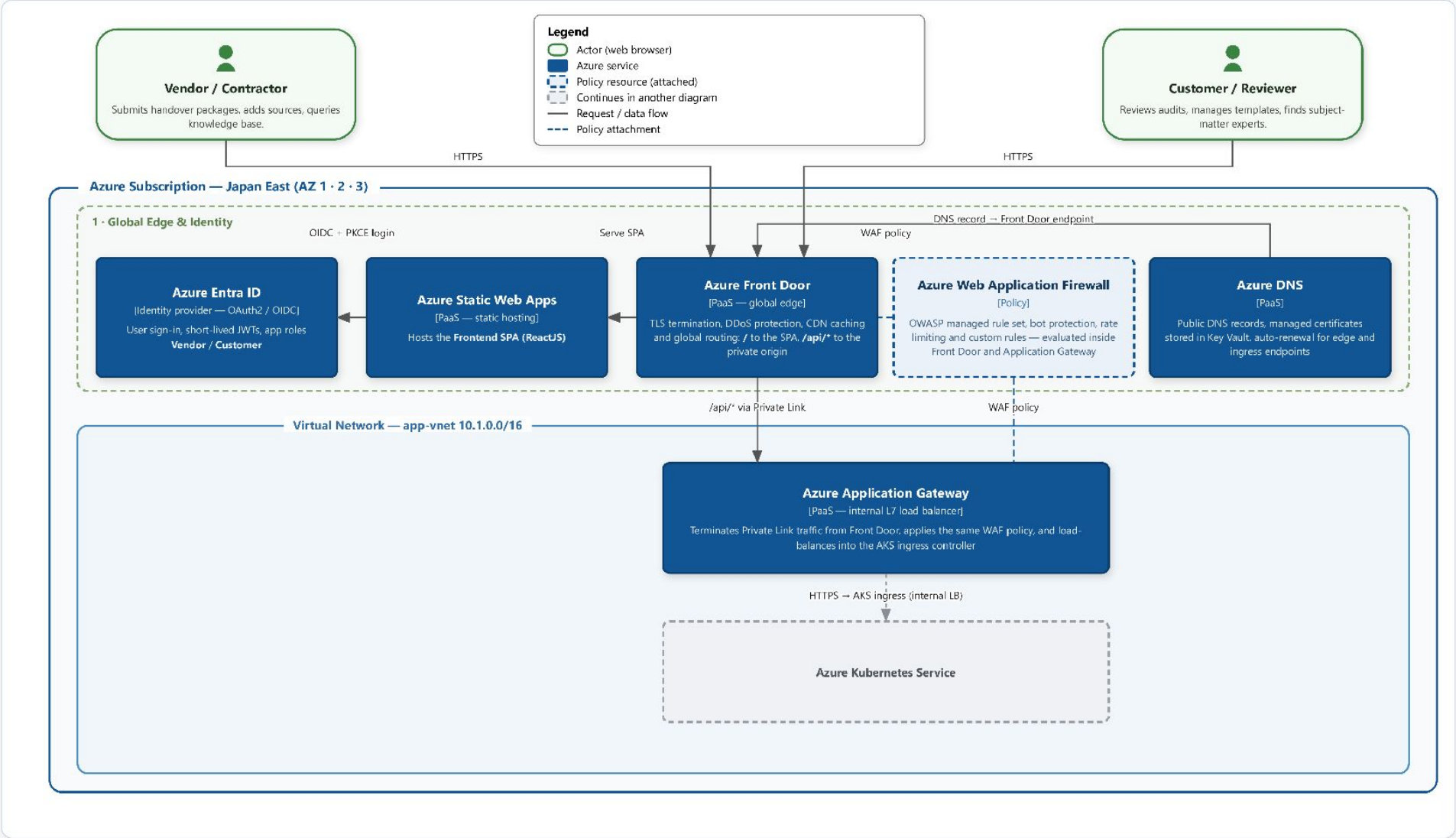
Container Diagram



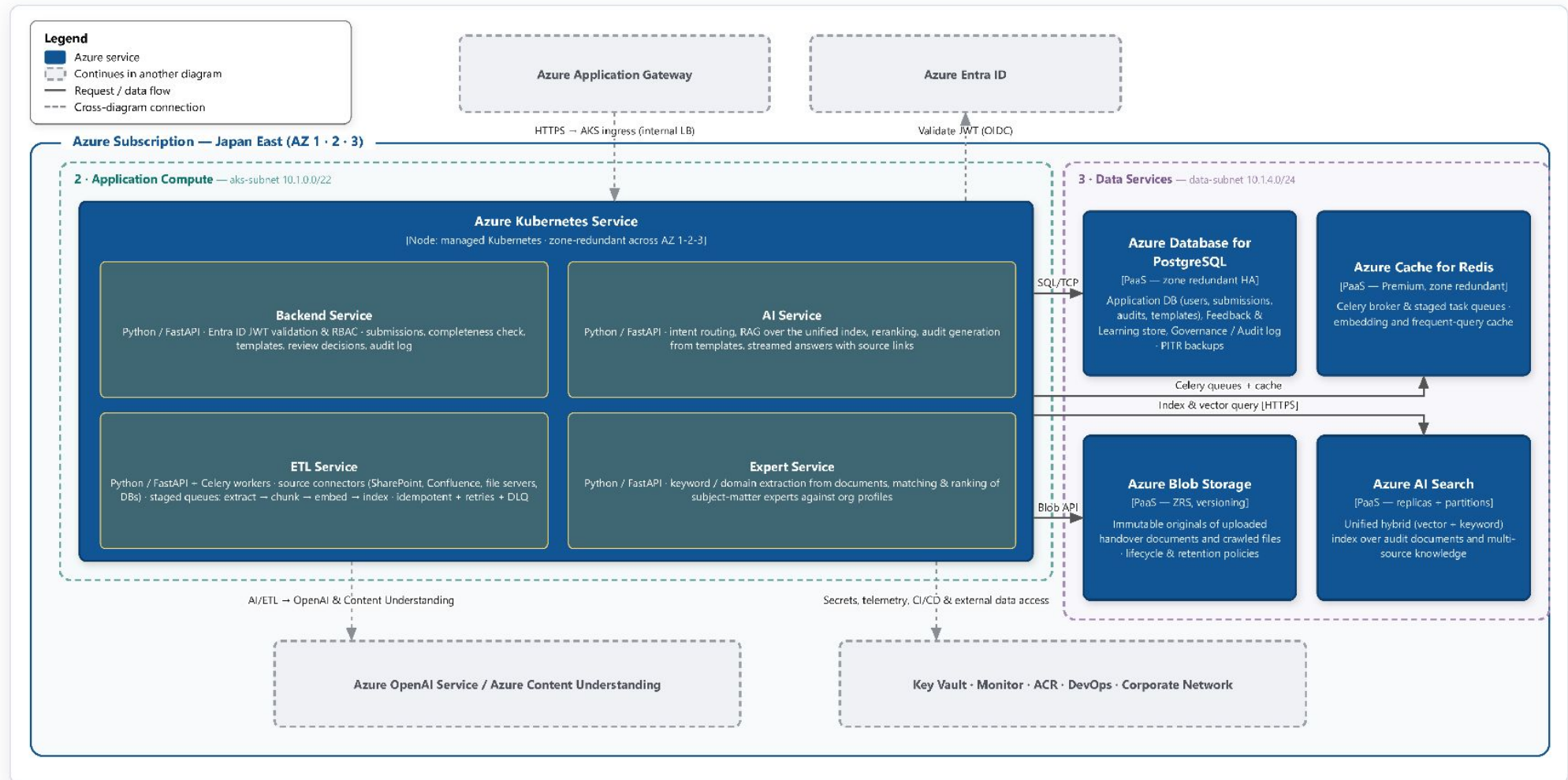
Infrastructure Architecture



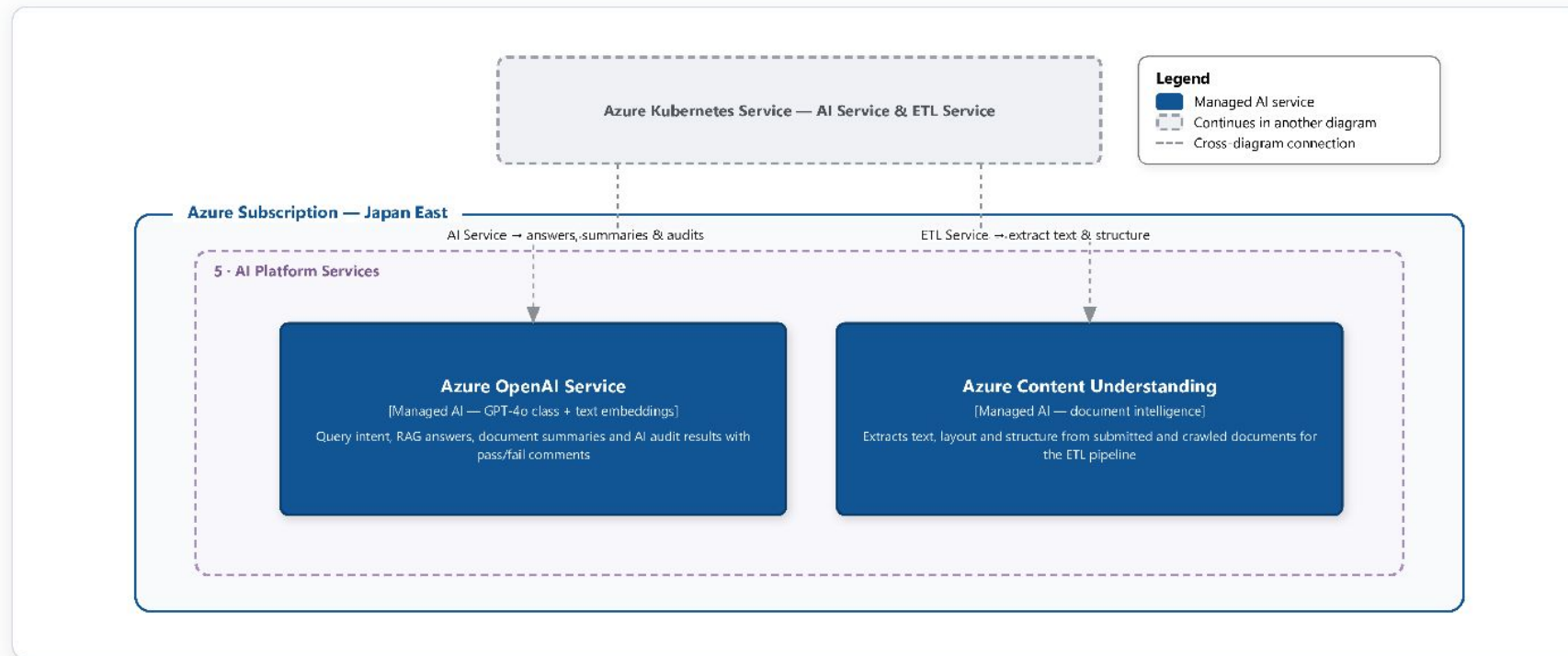
Infrastructure Architecture - Global Edge & Identity (1/4)



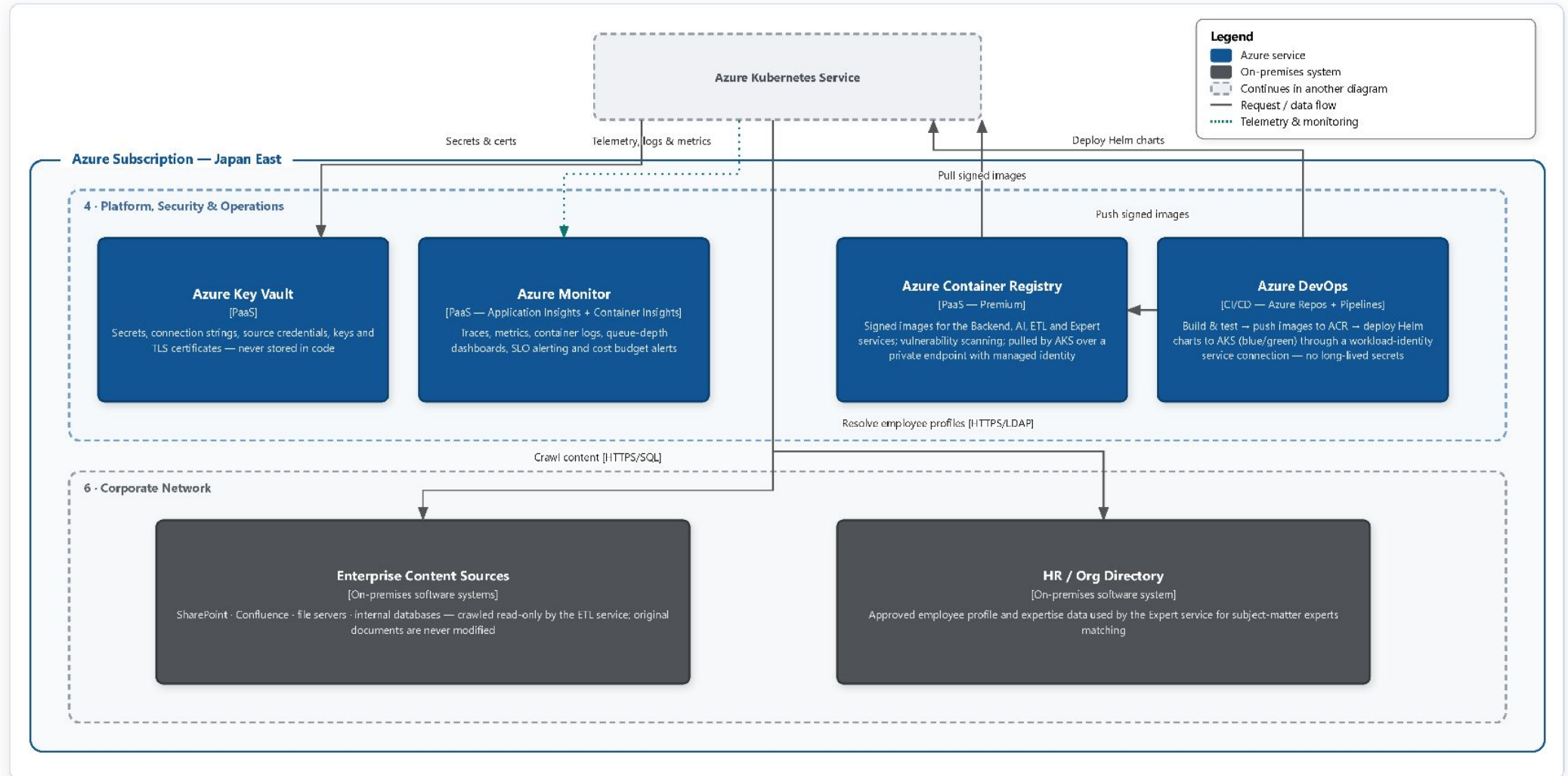
Infrastructure Architecture - Application Compute & Data Services (2/4)



Infrastructure Architecture - AI Platform Services (3/4)



Infrastructure Architecture - Platform, Security & Operations (4/4)



Security, Observability, and Scaling

Security

Entra ID and RBAC: Use Entra ID with OAuth2/OIDC sign-in. Short-lived JWTs are validated in every FastAPI dependency, and RBAC enforces Vendor and Customer scopes.

WAF-protected ingress: Apply one WAF policy across Front Door and Application Gateway, including OWASP managed rules, bot protection, rate limiting, and platform DDoS protection.

Private service connectivity: Keep application workloads in a protected private network and access documents, databases, AI services, and container images through private endpoints instead of the public Internet.

Protected data: Use TLS for data in transit, use Key Vault for secrets management, encrypt sensitive user fields at rest, and keep original documents immutable in Blob Storage under lifecycle and retention policies.

Observability

Centralized monitoring: Use Azure Monitor for traces, logs and metrics.

Operational dashboards: Track queue depth per ETL stage, audit duration, AI Search latency, token consumption, and service availability.

SLO and cost alerts: Alert when p95 query latency exceeds 3 seconds, an audit exceeds 10 minutes, the dead-letter queue is non-empty, or availability falls below 99.9%. Trigger budget alerts at 80%, 90%, and 100% of the monthly cap.

Auditable history: Store every submission, AI finding, and reviewer accept/reject decision in immutable PostgreSQL audit history with timestamps.

Scaling

Independent scaling: Scale the Backend, AI, and ETL services independently based on CPU, memory, queue depth, and request volume.

Burst handling: Add pods and workers automatically during submission spikes and high query concurrency without degrading latency.

Elastic search and cache: Expand Azure AI Search replicas/partitions and Redis capacity to maintain fast retrieval under peak load.

Cost-aware scaling: Maintain a lean baseline and scale out only when needed, then scale back down during normal operating periods.

Azure Fit

Capability	Service	Why
CDN	Azure Front Door	Global edge caching + TLS + DDoS + WAF
Identity	Azure Entra ID	Mandatory
SPA hosting	Azure Static Web Apps	Serverless
Load balancer	Azure Application Gateway	WAF + Keep AKS off the public Internet
Container platform	Azure Kubernetes Service	More configurable scaling
Relational database	Azure Database for PostgreSQL	Fully managed + HA + PITR
Cache & Celery broker	Azure Cache for Redis	Fully managed + HA
Object storage	Azure Blob Storage	ZRS + Versioning + Policies
Vector database	Azure AI Search	Fully managed + First-party integration with AOS & ACU
LLM & embeddings	Azure OpenAI Service	Enterprise-grade + Data staying inside Azure tenant boundary
Document extraction	Azure Content Understanding	Extract text, layout and structure from mixed document formats without custom OCR
Secrets management	Azure Key Vault	Central store for secrets, keys and certs
Observability	Azure Monitor	Traces, logs, metrics dashboards and alerts in one place
Registry & CI/CD	Azure Container Registry + Azure DevOps	Mandatory
Networking	VNet + Subnet + Private Link	Keep backend, data and AI services private

Sizing & Cost (1)

Workload sizing

Throughput

- 500 document submissions / hour
- 50 concurrent interactive queries

Latency

- First token < 2s (p95)
- Query P95 < 3s during bursts
- 95% of audits < 10min per package
- P99 end-to-end < 15 min

Resilience

- 99.9% monthly availability
- RTO < 30min · RPO < 15min · zero silent data loss

Platform footprint

- AKS zone-redundant AZ 1-2-3, four services (Backend, AI, ETL, Expert)
- AI Search sized by replicas (QPS) and partitions (index size)
- PostgreSQL zone-redundant HA + PITR
- Redis zone-redundant HA
- Blob Storage ZRS + Versioning

Development effort (1 PM, 2 Backend, 3 AIE, 2 Frontend, 2 QA)

Task	Man-Days
Project setup & cloud infrastructure	16
Authentication & authorization	8
Frontend application	56
Backend API	46
Document processing pipeline	34
AI search & knowledge base	26
Expert identification	10
Learning from interactions	10
Non-functional requirements	26
Testing	14
Deployment, documentation & training	18
Risk buffers	28
Total	304

Sizing & Cost (2)

Infrastructure cost — development (3 months)		
Component	Monthly	3 months
Azure OpenAI Service (dev + test)	900	2,700
Azure Database for PostgreSQL	150	450
Azure AI Search	120	360
Azure Monitor	90	270
CI/CD, test runner & support tooling	80	240
Azure Cache for Redis	70	210
Key Vault, private networking & DNS	60	180
Azure Blob Storage	35	105
Total (USD)	1,505	4,515

Infrastructure cost — production (steady state)	
Component	Monthly
Azure OpenAI Service (production inference)	2,500
Azure Database for PostgreSQL (HA + PITR)	350
Azure AI Search (standard tier)	350
Azure Monitor	200
Alerting, autoscale support & operational overhead	100
Azure Cache for Redis (HA)	150
Key Vault, private networking & DNS	150
Azure Blob Storage (ZRS + Versioning)	120
Total (USD)	3,920

Q & A

Thank You